

WICK, United Kingdom

WMO: 030750

Lat: 58.4539N

Lon: 3.0900W

Elev: 38

StdP: 100.86

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.1	-1.6	-5.4	2.4	-0.7	-4.1	2.7	0.5	17.9	4.6	16.1	4.5	3.1	260	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.4	18.2	15.1	17.1	14.3	16.1	13.7	15.8	17.5	15.0	16.4	14.3	15.6	5.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	10.7	16.5	14.2	10.2	15.6	13.7	9.8	15.0	44.4	17.7	42.1	16.4	40.3	15.7	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.9	13.2	11.7	DB	-5.7	21.7	2.3	1.3	-7.4	22.6	-8.7	23.3	-10.0	24.1	-11.7	25.0	(4)
(5)			WB	-6.2	17.7	2.2	0.9	-7.8	18.4	-9.1	18.9	-10.3	19.3	-12.0	20.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	4.1	4.0	5.0	6.7	8.4	11.0	12.9	13.0	11.7	9.2	6.4	4.4	(6)	
	DBStd	4.08	2.39	2.60	2.56	2.43	2.27	2.09	1.93	1.81	2.17	2.49	2.74	3.00	(7)	
	HDD10.0	1023	184	168	156	103	58	13	1	1	9	44	110	174	(8)	
	HDD18.3	3743	443	401	413	349	307	222	169	165	200	284	358	432	(9)	
	CDD10.0	321	0	0	1	3	10	42	91	94	59	19	2	1	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.8	6.7	6.7	6.6	6.0	5.2	5.0	4.5	4.7	5.4	6.1	6.4	6.2	(14)	
(15) Precipitation	PrecAvg	787	73	58	63	49	47	49	58	71	77	77	91	79	(15)	
	PrecMax	1006	140	135	117	96	81	106	99	151	160	176	168	138	(16)	
	PrecMin	562	23	15	14	13	10	12	24	17	24	20	32	28	(17)	
	PrecStd	109	29	26	24	21	19	21	21	34	28	34	38	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	12.1	13.0	14.9	16.4	18.3	20.6	19.9	18.9	15.6	13.0	11.9	(19)	
		MCWB	9.4	10.2	10.1	11.4	12.9	14.3	16.7	16.2	15.3	13.4	11.2	10.5	(20)	
	2%	DB	9.4	10.3	10.9	12.5	14.4	16.4	18.2	17.8	16.7	14.1	11.8	10.5	(21)	
		MCWB	8.2	8.9	8.8	9.8	11.5	13.3	15.3	14.9	13.9	12.2	10.4	9.3	(22)	
	5%	DB	8.4	9.0	9.7	11.2	13.0	15.2	17.1	16.8	15.6	13.2	11.1	9.6	(23)	
		MCWB	7.3	7.7	7.9	8.8	10.6	12.6	14.6	14.2	13.3	11.7	9.9	8.4	(24)	
	10%	DB	7.5	7.8	8.7	10.1	11.9	14.2	16.0	15.9	14.7	12.5	10.3	8.7	(25)	
		MCWB	6.5	6.5	7.1	8.2	9.9	12.0	13.9	13.7	12.8	11.2	9.3	7.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.6	10.6	12.0	13.7	15.2	17.4	16.7	15.7	14.0	11.7	10.7	(27)	
		MCDB	10.5	11.8	12.5	14.1	15.7	17.3	19.5	19.0	17.9	15.0	12.3	11.5	(28)	
	2%	WB	8.5	9.2	9.2	10.2	12.1	14.0	16.1	15.6	14.6	12.7	10.9	9.6	(29)	
		MCDB	9.2	10.1	10.6	12.0	13.8	15.8	17.7	17.2	16.1	13.5	11.5	10.4	(30)	
	5%	WB	7.5	7.9	8.2	9.2	11.0	13.2	15.1	14.8	13.8	12.1	10.2	8.6	(31)	
		MCDB	8.3	9.0	9.4	10.7	12.5	14.6	16.5	16.1	14.9	12.9	10.9	9.4	(32)	
	10%	WB	6.7	6.7	7.3	8.5	10.2	12.4	14.3	14.2	13.2	11.5	9.4	7.7	(33)	
		MCDB	7.5	7.6	8.4	9.8	11.6	13.8	15.6	15.3	14.3	12.4	10.2	8.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.7	5.3	5.4	5.8	5.4	5.3	5.4	5.3	4.9	4.4	4.4	(35)	
		MCDBR	4.8	5.6	6.5	6.6	6.8	6.7	6.7	6.3	6.0	5.2	4.6	5.1	(36)	
	5% WB	MCWBR	4.6	4.8	5.0	4.4	4.5	4.4	4.3	4.1	4.2	4.3	4.3	4.9	(37)	
		MCWBR	4.7	5.3	6.1	5.9	6.1	6.2	6.0	5.4	5.1	4.6	4.5	5.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.317	0.340	0.360	0.379	0.372	0.369	0.376	0.365	0.353	0.340	0.319	0.300	(40)	
		taud	2.194	2.303	2.330	2.315	2.389	2.429	2.421	2.458	2.484	2.408	2.242	2.131	(41)	
	Edn at Noon	Edn at Noon	491	680	779	826	860	868	851	836	788	676	487	377	(42)	
		Edh at Noon	50	71	91	108	107	105	103	93	78	64	47	38	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.36	0.97	2.05	3.54	4.85	4.77	4.52	3.62	2.42	1.25	0.49	0.23	(44)	
	RadStd	RadStd	0.04	0.10	0.15	0.29	0.29	0.34	0.30	0.23	0.20	0.10	0.04	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page